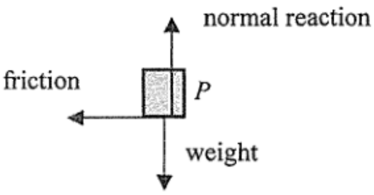
Q1

1. (a) conduction / radiation	1A	
	1	
(b) Will not, as the silvery surface of aluminum foil is a poor radiation absorber / good radiation reflector.	1A+1A	
	2	
(c) (i) The water droplets come from the moisture / water vapour in the air (surroundings).	1A	
	1	
(ii) $E = (0.40 \times 10^{-3})(2.26 \times 10^{6})$ $= 904$ J	1M 1A	
	2	
2. (a) (i) $E_K = \frac{1}{2}\left(6.63\times10^{-26}\right)(500)^2$ $= 8.2875 \times 10^{-21}$ J $\approx 8.29 \times 10^{-21}$ J	1M 1A	Accept: $(8.29 \sim 8.30) \times 10^{-21}$ J
	2	
(ii) $E_K = \frac{3RT_0}{2N_A}$ $2\left(8.29\times10^{-21}\right)$		

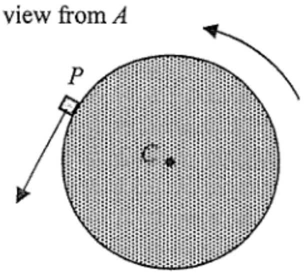
Q2

$= 400.247\text{ (K)} \approx 400\text{ (K)}$	1A	
	2	
(b) As temperature remains at T_0,		
E_K remains unchanged / $c_{r.m.s.}$ depends on temperature,	1A	
so root-mean-square speed $c_{r.m.s.}$ remains unchanged.	1A	
	2	

Q3

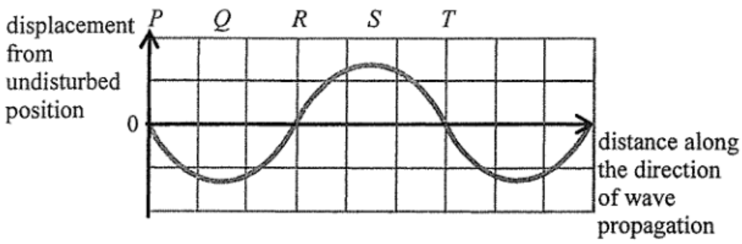
3. (a)	By $Fs = \frac{1}{2}mv^2$,		Or $v^2 = u^2 + 2as$ and $F = ma$
	$0.30 \times 0.75 = \frac{1}{2}(0.20)\, v^2$	1M	$v^2 = 0^2 + 2(\frac{0.30}{0.20})(0.75)$
	$v = 1.5\ \text{m s}^{-1}$	1A	
		2	
(b)		2A	Accept: normal force, gravity
		2	
(c)	Height $= \frac{1}{2}gt^2 = \frac{1}{2}(9.81)(0.35)^2$	1M	
	$= 0.6008625\ \text{m} \approx 0.601\ \text{m}$	1A	Accept: 0.60 m ~ 0.613 m
		2	

Q4

4.	(a)	(i)	$F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$ $= 0.36$ N	1M 1A	
				2	
		(ii)	Top view from A  	1A	
				1	
	(b)	(i)	same angular speed	1A	
				1	
		(ii)	different / smaller magnitude of centripetal acceleration ($a_Q < a_P$) as Q takes a different / smaller radius ($r_Q < r_P$)	1A 1A	
				2	

Q5

5. (a)



5. (a) (graph above)	2A	
	2	
(b) (i) *P* / *R* / *T*	1A	
	1	
(ii) *R*	1A	
	1	

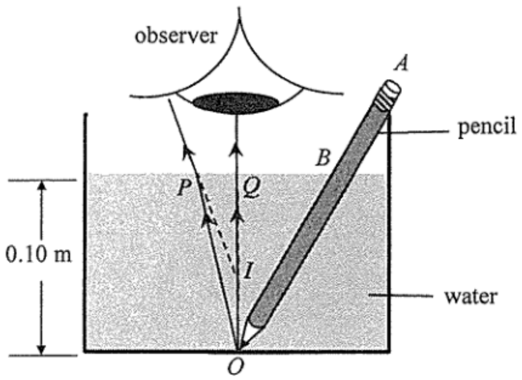
Q6

6.	(a)	Apply $d \sin\theta = m\lambda$, where θ is the deviation angle of the diffracted beam, m is the order of diffraction maximum. $\tan\theta = \frac{x/2}{L} \Rightarrow \theta = 24.702430°$ or $\sin\theta = \frac{x/2}{\sqrt{L^2+(x/2)^2}} = 0.417906$	1M		
		$d = \frac{\lambda}{\sin\theta} = \frac{650 \times 10^{-9}}{0.418}$	1M		
		$= 1.555375 \times 10^{-6}$ m ≈ 1.56 μm	1A		Accept: 1.56 μm ~ 1.6 μm
				3	
	(b)	x will decrease as λ decreases, $\sin\theta$ and θ will decrease.	1A 1A		
				2	

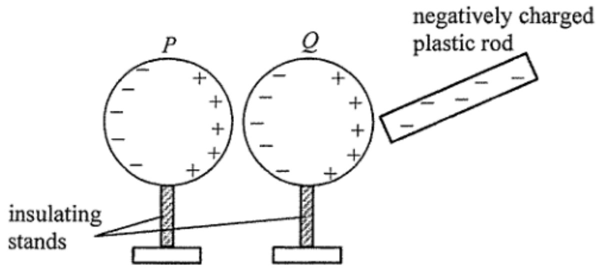
Q7

7.	(a)	The speed of sound increases (linearly) with air temperature. <u>Or</u> The higher the air temperature, the greater is the speed of sound.	1A		
				1	
	(b)	The air near the ground is cooler while the air layers higher above are relatively warmer,	1A		
		the sound wave bends downward (towards the normal) as it travels slower when going from a higher temperature (upper) layer into a low temperature (lower) layer and vice versa, i.e. refraction occurs.	1A		
				2	
	(c)	(i) Speed ratio $\frac{3.00\times10^{8}}{337} = 8.90208 \times 10^{5} \approx 8.90 \times 10^{5}$	1A		
				1	
		(ii) The distance is given by 337×3.0	1M		e.c.f. from (c)(i) using incorrect sound speed
		$= 1011$ m	1A		Accept: 1010 m ~ 1011 m
				2	

Q8

8. (a) 	1A 1M		1M for image I formed by the two refracted rays.
		2	
(b) $1.33 = \dfrac{\sin r}{\sin 1.5^\circ}$ $r = 1.995175^\circ \approx 2.00^\circ$	1M 1A		
		2	
(c) $\dfrac{h_1}{h} = \dfrac{\tan 1.5^\circ}{\tan 2.00^\circ} = 0.751681$ (or $\tan 1.5^\circ = \frac{PQ}{0.10}$ & $\tan 2^\circ = \frac{PQ}{h_1}$) $\Rightarrow h_1 \approx 0.075$ m (i.e. 7.5 cm)	1M 1A		e.c.f. from (b), explicit step / method shown Accept: $\dfrac{\text{real depth}}{\text{apparent depth}} = \dfrac{h}{h_1} = n$
		2	

Q9

9. (a) 	2A	
	2	
(b) No, it remains unchanged as charging is through induction (without any contact or transfer of charges).	1A+1A	
	2	
(c) Hold / bring the unknown charged rod (assume positive) near sphere *P*, which carries positive charges, without touching it.	1A	
If *P* is repelled by the rod, the rod should carry positive charges. <u>And / Or</u> If *P* is attracted by the rod, the rod should carry negative charges.	1M	1M for the correct procedure to identify the polarity of the charged rod.
	2	

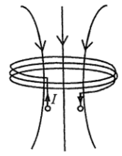
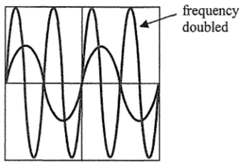
Q10

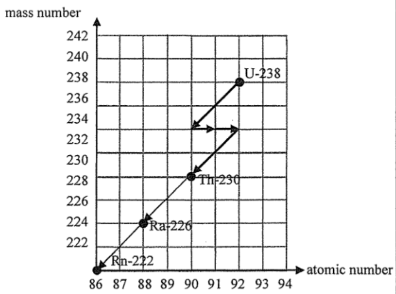
10. (a) 1. Use the protractor to measure or mark a specific angle at which the ball is released. Make sure the angle is measurable and recorded.	1A
2. Hold the ball at the marked angle and release it from rest with the string taut, allowing it to swing like a pendulum.	1A
3. Measure the angle at which the ball reaches its highest point and record the value. Or Observe the angle at which the ball swings back to the original releasing position.	1A
4. Compare the measurements of the ball's initial angle of release to the angle it reaches on the opposite side of the swing. (The angles equal implies total mechanical energy is conserved.)	1A
	4
(b) (i) (I) The tension forces are vertical / perpendicular to the direction of collision, the condition of no external forces along the colliding / horizontal direction is still fulfilled, the law can be applied. Or The tensions in the string acting on the spheres are balanced by the weights of the spheres so that the net force acting on the system is zero.	1A
	1
(II) By conservation of total momentum, v_E is $2 \times 0.50 = 1.0$ m s^{-1}	1A
However, the final total kinetic energy would then be $\frac{1}{2}(0.020)(1.0)^2 = 0.01\,\text{J} > \frac{1}{2}(2\times0.020)(0.50)^2 = 5\times10^{-3} = 0.005\,\text{J}$	1M
which is greater than the total kinetic energy before collision, thus it is not possible.	1A
	3
(ii) Not perfectly elastic as some kinetic energy is lost as sound / thermal energy, the total kinetic energy is not conserved.	1A+1A

Q11

11. (a) As the foil becomes longer / l increases while its cross-sectional area becomes smaller / A decreases due to decrease in width w,	1A	
its resistance would increase according to $R = \frac{\rho l}{A}$.	1A	
	2	
(b) (i) $V_{in} = I_1(R_1 + R_2)$	1A	
	1	
(ii) Before: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{500}{470+500} - \frac{470}{470+470} = 0.0154639$	1M	
After: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{501}{470+501} - \frac{470}{470+470} = 0.0159629$		
Thus $\frac{V_{out}}{V_{in}}$ increases 3.22687% ≈ 3.23%.	1A	Accept ±(3.2 ~ 3.23)% as the change
	2	
(iii) (I) crack ①	1A	
(II) along AB	1A	
	2	

Q12

12. (a) 	2A	1A for correct direction (also accept field lines pointing upwards for observers with different perception of current direction) 1A for correct field pattern
	2	
(b) (i) The a.c. flowing in transmitter coil T generates a changing magnetic field,	1A	
thus by electromagnetic induction an induced e.m.f. is produced in receiver coil R to oppose the changing magnetic flux experienced by it.	1A	
	2	
(ii) CRO display 	2A	1A for correct frequency 1A for correct amplitude
	2	
(c) If the back cover is metallic, eddy currents will be produced (by induction),	1A	
loss of energy / heating effect results / magnetic field / flux is blocked by the metal case and thus making wireless charging impossible. Or Non-metallic materials such as glass are insulators, and no eddy currents are produced. As a result, energy loss / heating effect will be minimized / the magnetic field can pass through the back cover easily.	1A	
	2	

Solution	Marks	Remarks
13. (a) (i) $k = \frac{\ln 2}{t_{½}}$		
$= \frac{\ln 2}{3.82 \times 24 \times 3600}$	1M	
$= 2.1001405 \times 10^{-6}\ \text{s}^{-1} \approx 2.10 \times 10^{-6}\ (\text{s}^{-1})$	1A	
	2	
(ii) $A = kN$		
$N = \frac{48}{2.10 \times 10^{-6}}$	1M	e.c.f. from (a)(i)
$= 2.285561 \times 10^{7} \approx 2.29 \times 10^{7}$	1A	Accept: $(2.28 \sim 2.3) \times 10^{7}$
	2	
(iii) As radon is a gas that can be inhaled into the lungs of human,	1A	
the relatively strong ionizing power of α particles emitted by decaying radon / radioactive gas might affect the organs / cells nearby.	1A	
	2	
(b) 	2A	
	2	